

Does Welsh Media Need a Review?

Using ML to Detect Bias in Political Reporting

Cai Parry-Jones · D200 Machine Learning in Economics · March 2026

Motivation

Reform calls for review into BBC and Plaid relationship



Rhuanedd Richards, became director of Wales in 2021, and took up the interim position in June 2025

Gareth Lewis, Wales political editor and **Gwyn Loader**, Newyddion 54C chief correspondent

11 November 2025
Updated 12 November 2025

Reform UK has called for a "review into the relationship" between the BBC and Plaid Cymru.

BBC News, Nov 2025

NEWS

Tory MS raises concerns over 'impartial' media coverage of Senedd election

15 Jan 2026 4 minute read



Conservative Senedd Member for Brecon and Radnorshire, James Evans.

Emily Price

A Welsh Conservative Senedd Member has raised strong concerns about a lack of "fair, balanced and impartial" media coverage in the run up to the most significant election since the dawn of devolution.

Nation.Cymru, Jan 2026

*Growing accusations of bias in Welsh media by political parties.
Fair criticism, or empty rhetoric?*

Literature

Statement bias is a powerful form of media bias (D'Alessio & Allen, 2000; Eberl et al., 2017)

NLP can detect media bias (Rodrigo-Gines et al., 2024; Spinde et al., 2023)

LLMs beginning to exceed fine-tuned classifiers on sentiment tasks (Shao, 2025)

Model cascading: cheap model first, costly LLM where needed (Yue et al., 2024)

Data & Scope

Source

Nation.Cymru

Award winning* Welsh political news outlet

Data access: WordPress REST API

11,847 articles (2017–2026)

Filtered by politics keywords

Analysis Scope

Only one news outlet

Primary:

Reform UK vs Plaid Cymru

15,583 mentions · News only · 2022-2026

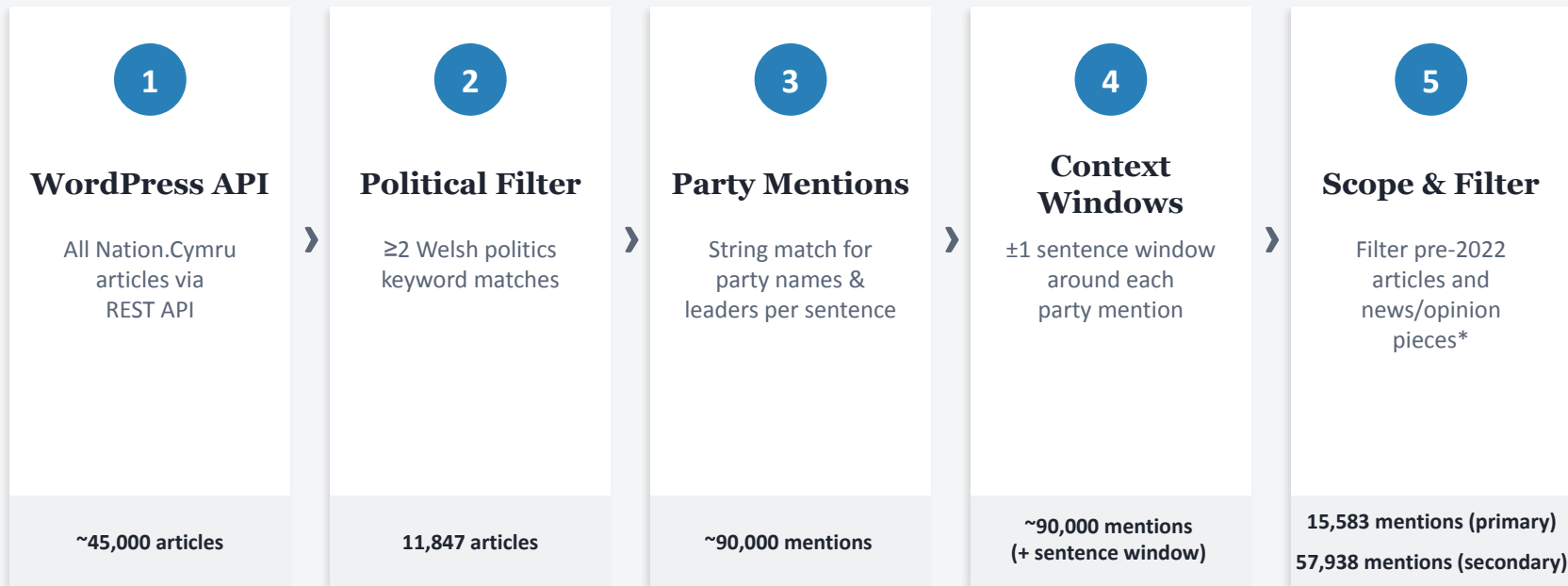
Secondary:

4 parties · News + Opinion · 2022-2026

57,938 mentions

** Online News Site of the Year, Wales Media Awards 2025, 17 January 2026.*

Data Pipeline



** Only news articles for primary analysis. Secondary includes opinion articles too.*

Two-Stage ML Pipeline

Stage 1: NLP Bias Detection

himel7/bias-detector

(RoBERTa, fine-tuned on BABE)

Classifies ± 1 sentence context around each party mention as biased / non-biased

Runs locally



Stage 2: LLM Attribution

Claude Sonnet 4.6

(Anthropic API)

Determines: is bias directed at the party, or is the party just the speaker? How negative/positive was the mention?

Returns: on_target, sentiment score, reasoning

~\$11 total API cost

Stage 1: RoBERTa Bias Detector

Examples

Mention: “Reform’s comments show they are not fit to govern.”

RoBERTa output. Bias confidence: 0.955

Mention: “It was Plaid Cymru that stepped in to rescue local government from a catastrophic Labour draft budget that would have pushed councils towards bankruptcy, mass job losses and double-digit council tax rises,” he said.

RoBERTa output. Bias confidence: 0.830

Stage 2: LLM Prompted Classifier Examples

Mention: “Reform’s comments show they are not fit to govern.”

LLM output. Reasoning: Reform is described as 'not fit to govern' — a direct negative judgement of their competence.

on_target: 1

sentiment_score: -1.0 (very negative)

Mention: “It was Plaid Cymru that stepped in to rescue local government from a catastrophic Labour draft budget that would have pushed councils towards bankruptcy, mass job losses and double-digit council tax rises,” he said.

LLM output. Reasoning: Plaid is positively framed — 'Plaid Cymru stepped in to rescue local government from a catastrophic Labour draft budget'.

on_target: 1

sentiment_score: 0.5 (positive)

Why LLM Matters

33.4%

of biased mentions
classified as off-target

5-point

categorical sentiment
scale per mention

~6%

wrong-party
classification rate

Off-target, wrong-party example: “Threatening to withhold funding while showing zero grasp of the institution’s origins or philosophical tradition only underlines why the far right is a threat to our Welsh way of life.” Plaid Cymru councillor Gareth Parry, from Bangor, wrote: “@ZiaYusufUK threatening Welsh universities for their student debating socs making their own decisions?”

LLM output. Reasoning: Plaid Cymru councillor is speaking/quoting, criticising Reform's actions. Plaid is the speaker, not the subject of judgement.

on_target: 0

sentiment_score: 0.0 (neutral)

Primary Results: Measurable Differential Framing

2×

higher on-target
bias rate for Reform

-0.7 vs -0.2

Reform vs Plaid mean
sentiment score

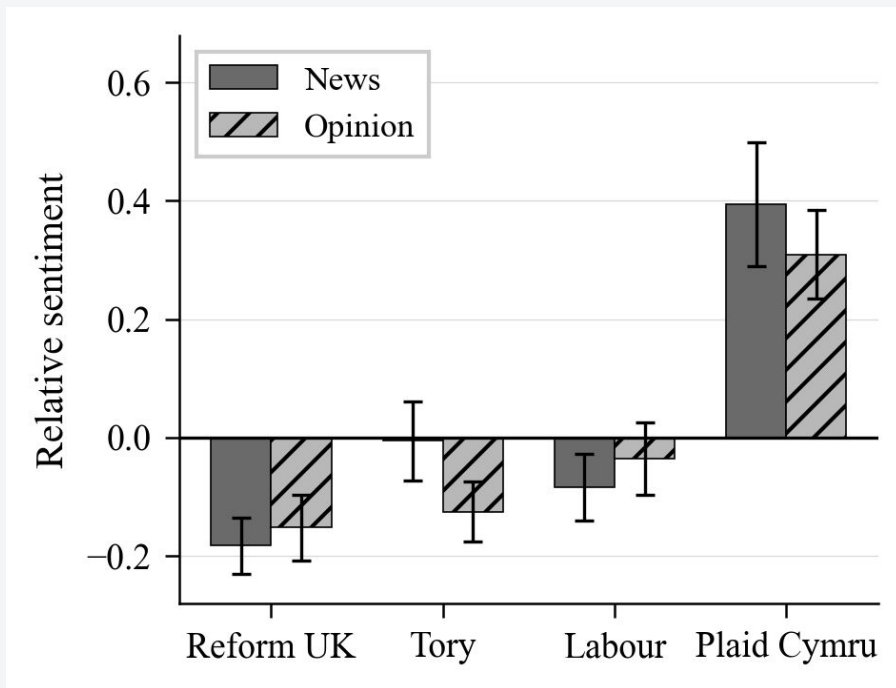
5×

higher strong negative
sentiment mentions for Reform

Bias rate z-test: $z = 15.75$, $p < 0.001$, Cohen's $h = 0.25$

Sentiment t-test: $t = -20.93$, $p < 0.001$, Cohen's $d = -1.04$

Secondary Results: Plaid Cymru Is the Outlier



Mean on-target sentiment relative to article-type baseline. Error bars: 95% CI.

Limitations & Next Steps

Limitations

- Single outlet, no gatekeeping or coverage bias detection, as well as no analysis of Wales media more broadly
- Keyword matching misses coreference (anaphoric references)
- LLM stochastic; ~6% wrong-party rate
- Mentions treated as independent (within-article correlation)
- 250-mention sampling cap in secondary analysis



With More Time

- Extend to BBC Wales and WalesOnline
- Measure gatekeeping and coverage bias across outlets
- Introduce ML into preprocessing data pipeline for political party labelling
- Cluster standard errors by article
- Remove 250-mention sampling cap
- Further analysis: Time series, author, images

Thank You

github.com/caitpi/d200_media_bias

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